Direct methods

- Gaussian elimination
- LU factorization
- Cholesky factorization

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1. Find LU factorization of $\begin{bmatrix} 2 & 1 & 1 \\ 4 & -6 & 0 \\ -2 & 7 & 2 \end{bmatrix}$
   - using Doolittle's factorization
   - using Crout's factorization

   via possible elementary row operations.

2. Solve this system of equations $\begin{bmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$ using
   - Gaussian elimination method
   - Doolittle factorization
   - Crout's method
   - Cholesky decomposition.

3. Find the inverse of
$$S = \begin{bmatrix} 2 & 4 & -2 \\ 4 & 9 & -3 \\ -2 & -3 & 7 \end{bmatrix}$$
   using Gaussian elimination.

Programming assignment

4. Write a computer program (using your preferred language) to solve a system of equations (whenever applicable) using
   - Gaussian elimination
   - LU decomposition
   - Cholesky decomposition.

5. (Advanced/optional) : Try for parallel implementation of above mentioned algorithms on your preferred parallel architecture.